

Chapter 3 Test model

Multiple Choice Questions (70 Questions)

1. In the ER model, an entity that does not have a key attribute and depends on another entity for its identity is called a:
 - a. Strong entity
 - b. Composite entity
 - c. Weak entity
 - d. Recursive entity
-

2. Which of the following is NOT a type of attribute in the ER model?
 - a. Simple
 - b. Composite
 - c. Recursive
 - d. Multivalued
-

3. The attribute "Address" with subcomponents like Street, City, and ZipCode is best classified as:
 - a. Simple
 - b. Derived
 - c. Composite
 - d. Multivalued
-

4. In ER diagrams, multivalued attributes are represented by:
 - a. Single oval
 - b. Double oval
 - c. Dashed oval
 - d. Rectangular box
-

5. A key attribute in an ER diagram is typically:
 - a. Bolded
 - b. Italicized

- c. Underlined
 - d. Enclosed in brackets
-

6. Which relationship type in the COMPANY database is recursive?
- a. WORKS_FOR
 - b. MANAGES
 - c. SUPERVISION
 - d. DEPENDENTS_OF
-

7. The degree of a relationship type refers to:
- a. The number of attributes in the relationship
 - b. The cardinality ratio between entities
 - c. The number of participating entity types
 - d. The total number of instances in the relationship set
-

8. In the COMPANY schema, the relationship between EMPLOYEE and PROJECT is:
- a. 1:1
 - b. 1:N
 - c. N:1
 - d. M:N
-

9. What uniquely identifies a weak entity?
- a. Its own primary key
 - b. A foreign key from another table
 - c. A combination of its partial key and the owner entity's key
 - d. A derived attribute
-

10. The relationship "MANAGES" in the COMPANY database has which cardinality from DEPARTMENT to EMPLOYEE?
- a. (0,1)
 - b. (1,1)
 - c. (0,N)
 - d. (1,N)
-

11. In ER modeling, the term "entity set" refers to:
- a. A collection of relationship instances
 - b. The schema definition of an entity type
 - c. The current collection of entity instances of a given type
 - d. A set of composite attributes
-

12. Which of the following best describes a value set?
- a. A list of all entity instances
 - b. The domain or data type associated with an attribute
 - c. The set of all possible relationships
 - d. The set of key attributes for an entity
-

13. An attribute like "HoursPerWeek" in the WORKS_ON relationship is an example of:
- a. A derived attribute of EMPLOYEE
 - b. An attribute of the relationship type
 - c. A composite attribute of PROJECT
 - d. A partial key
-

14. In a recursive relationship, role names are necessary to:
- a. Identify the weak entity
 - b. Distinguish between different participations of the same entity type
 - c. Define cardinality constraints
 - d. Denote multivalued attributes
-

15. Which of the following is a correct statement about key attributes?
- a. Only one key attribute can exist per entity type
 - b. Key attributes must always be simple
 - c. An entity type may have multiple keys
 - d. Keys cannot be composite
-

16. The participation of DEPARTMENT in the WORKS_FOR relationship is:
- a. Total
 - b. Partial
 - c. Mandatory for all departments
 - d. (1,1)

17. In the (min, max) notation, (1,1) for EMPLOYEE in WORKS_FOR implies:

- a. An employee may work for zero or one department
 - b. An employee must work for exactly one department
 - c. An employee can work for many departments
 - d. A department must have at least one employee
-

18. Which of the following relationships in the COMPANY database is 1:N?

- a. WORKS_ON
 - b. SUPERVISION
 - c. MANAGES
 - d. DEPENDENTS_OF
-

19. The identifying relationship for the DEPENDENT entity type is:

- a. WORKS_FOR
 - b. DEPENDENTS_OF
 - c. SUPERVISION
 - d. CONTROLS
-

20. In ER diagrams, relationship types are represented by:

- a. Rectangles
 - b. Ovals
 - c. Diamonds
 - d. Double rectangles
-

21. Which of the following is true about composite multivalued attributes?

- a. They cannot exist in ER modeling
 - b. They are denoted as {Attribute(Component1, Component2, ...)}
 - c. They are always key attributes
 - d. They are represented by single ovals
-

22. The attribute SSN in the EMPLOYEE entity type serves as a:

- a. Partial key
- b. Composite attribute

- c. Key attribute
 - d. Derived attribute
-

23. If an entity type E has an attribute A defined as $A: E \rightarrow P(V)$, then A is:
- a. Always simple
 - b. Always derived
 - c. Possibly multivalued
 - d. Necessarily composite
-

24. Which of the following is NOT a valid reason to model a concept as a relationship instead of an attribute?
- a. It connects two entity types
 - b. It has its own descriptive attributes
 - c. It represents a property of a single entity
 - d. It involves multiple entities
-

25. In the COMPANY database, the start date of a department manager is best modeled as:
- a. An attribute of DEPARTMENT
 - b. An attribute of EMPLOYEE
 - c. An attribute of the MANAGES relationship
 - d. A derived attribute
-

26. Total participation of an entity type in a relationship is shown in ER diagrams by:
- a. A single line
 - b. A double line
 - c. An arrowhead
 - d. A dashed box
-

27. The relationship "CONTROLS" in the COMPANY schema is between:
- a. EMPLOYEE and PROJECT
 - b. DEPARTMENT and PROJECT
 - c. EMPLOYEE and DEPARTMENT
 - d. DEPENDENT and EMPLOYEE
-

28. Which of the following statements about weak entities is FALSE?
- a. They must participate in an identifying relationship
 - b. They have a partial key
 - c. They can exist independently of their owner entity
 - d. Their identifying relationship is typically total
-
29. In the (min, max) notation, what does (0,n) mean?
- a. Mandatory participation with at most n instances
 - b. Optional participation with no upper limit
 - c. Exactly n instances required
 - d. At least one and at most n instances
-
30. The entity type PROJECT in the COMPANY database has which of the following as a key?
- a. Location
 - b. Department number
 - c. Project number
 - d. Manager SSN
-
31. Which of the following best defines a relationship instance?
- a. A template for relationships
 - b. A specific association between entity instances
 - c. A set of all possible relationships
 - d. A type of attribute
-
32. Role names in a recursive relationship are used to:
- a. Replace key attributes
 - b. Indicate the direction of cardinality
 - c. Differentiate the two participations of the same entity
 - d. Denote weak entities
-
33. The number of hours an employee works on a project is stored as:
- a. An attribute of EMPLOYEE
 - b. An attribute of PROJECT

- c. An attribute of the WORKS_ON relationship
 - d. A composite attribute of DEPARTMENT
-

34. Which of the following is a correct ER modeling principle?
- a. Every attribute must be simple
 - b. Relationships cannot have attributes
 - c. A relationship type can exist between more than two entities
 - d. Weak entities must have a full key
-

35. In the COMPANY schema, an employee's supervisor is represented using:
- a. A foreign key in the EMPLOYEE table
 - b. The SUPERVISION recursive relationship
 - c. A multivalued attribute
 - d. A derived attribute
-

36. The partial key of a weak entity is:
- a. Sufficient to uniquely identify the entity alone
 - b. Combined with the owner's key to form a full identifier
 - c. Always a composite attribute
 - d. Never underlined
-

37. Which notation is used to represent a weak entity in an ER diagram?
- a. Double rectangle
 - b. Dashed rectangle
 - c. Rectangle with rounded corners
 - d. Shaded rectangle
-

38. The value set for an attribute is analogous to what in programming?
- a. Variable name
 - b. Function
 - c. Data type
 - d. Loop structure
-

39. In the MANAGES relationship, if each department must have exactly one manager, the participation of DEPARTMENT is:
- a. (0,1)
 - b. (1,1)
 - c. (0,N)
 - d. (1,N)
-

40. Which of the following is a binary relationship?
- a. A relationship among three entity types
 - b. A relationship between two instances of the same entity type
 - c. A relationship with a degree of 2
 - d. Both b and c
-

41. An entity type may have more than one key. In ER diagrams, how are they shown?
- a. Only the primary key is underlined
 - b. All keys are underlined
 - c. Keys are bolded
 - d. Keys are listed in a separate legend
-

42. The attribute "PreviousDegrees" for a STUDENT, with components like College and Year, is best modeled as:
- a. Simple attribute
 - b. Composite attribute
 - c. Multivalued composite attribute
 - d. Derived attribute
-

43. In ER modeling, existence dependency implies:
- a. Optional participation
 - b. Total participation
 - c. Recursive relationship
 - d. Weak entity without a partial key
-

44. Which of the following relationships has a descriptive attribute in the COMPANY database?
- a. WORKS_FOR

- b. MANAGES
 - c. WORKS_ON
 - d. SUPERVISION
-

45. The relationship set is:
- a. The schema for a relationship
 - b. The current collection of relationship instances
 - c. The set of all possible entity types
 - d. Synonymous with relationship type
-

46. In the (min, max) notation, the numbers are read:
- a. From the relationship toward the entity
 - b. From the entity toward the relationship
 - c. Left to right only
 - d. Based on alphabetical order
-

47. Which of the following is NOT true about the DEPENDENT entity?
- a. It is a weak entity
 - b. It is identified via the DEPENDENTS_OF relationship
 - c. Its partial key is Name
 - d. It has a full key attribute like SSN
-

48. A relationship type of degree 3 is called:
- a. Binary
 - b. Ternary
 - c. Recursive
 - d. Weak
-

49. In the COMPANY ER diagram, the attribute "Sex" is:
- a. Composite
 - b. Multivalued
 - c. Simple
 - d. Derived
-

50. The "Start Date" of a department manager is associated with which relationship?
- a. WORKS_FOR
 - b. MANAGES
 - c. SUPERVISION
 - d. CONTROLS
-

51. Which of the following best describes the purpose of conceptual database design?
- a. To write SQL queries
 - b. To design the physical storage structure
 - c. To create a high-level representation of data requirements
 - d. To optimize query performance
-

52. UML Class Diagrams are used in industry as an alternative to:
- a. Relational algebra
 - b. ER diagrams
 - c. SQL
 - d. Normalization
-

53. In ER diagrams, attributes are connected to entity types using:
- a. Dashed lines
 - b. Arrows
 - c. Straight lines
 - d. Curved lines
-

54. The entity type CAR having both VIN and (Number, State) as keys illustrates that:
- a. Keys must be simple
 - b. An entity type can have multiple keys
 - c. Only one key can be underlined
 - d. Composite keys are invalid
-

55. The power set $P(V)$ in the attribute function $A: E \rightarrow P(V)$ allows for:
- a. Only single values
 - b. Only composite values
 - c. Multivalued attributes
 - d. Derived attributes only

56. In a recursive relationship like SUPERVISION, the cardinality from supervisor to supervisee is typically:

- a. 1:1
 - b. 1:N
 - c. N:1
 - d. M:N
-

57. Which of the following is a correct statement about total participation?

- a. It is shown with a single line
 - b. It means every entity instance must participate in at least one relationship instance
 - c. It applies only to weak entities
 - d. It is optional
-

58. The relationship "WORKS_FOR" in the COMPANY database has what cardinality?

- a. 1:1
 - b. 1:N
 - c. N:1
 - d. M:N
-

59. In ER modeling, a derived attribute is:

- a. Stored in the database
 - b. Computed from other attributes
 - c. Always multivalued
 - d. Represented by a double oval
-

60. Which of the following is NOT part of the initial entity types identified for the COMPANY database?

- a. EMPLOYEE
 - b. DEPARTMENT
 - c. LOCATION
 - d. DEPENDENT
-

61. The attribute "Name" for DEPENDENT serves as a:

- a. Full key
 - b. Foreign key
 - c. Partial key
 - d. Composite key
-

62. In the ER model, a relationship type is conceptually similar to a:

- a. Table in the relational model
 - b. Primary key
 - c. Domain
 - d. Tuple
-

63. The (min, max) constraint (0,1) for EMPLOYEE in MANAGES means:

- a. Every employee manages a department
 - b. An employee may manage at most one department
 - c. Each department has exactly one manager
 - d. No employee can manage a department
-

64. Which of the following relationships is NOT binary?

- a. WORKS_ON
 - b. SUPERVISION
 - c. A hypothetical relationship among EMPLOYEE, DEPARTMENT, and PROJECT
 - d. MANAGES
-

65. In ER diagrams, a weak entity type is connected to its identifying relationship by:

- a. A single line
 - b. A double line
 - c. A dashed line
 - d. An arrow
-

66. The term "entity" in the ER model refers to:

- a. A table
- b. A row in a table
- c. A real-world object with independent existence
- d. A column

-
67. Which of the following attributes would most likely be multivalued?
- a. SSN
 - b. BirthDate
 - c. CollegeDegrees
 - d. Sex
-
68. The process of grouping entities with the same attributes into a type is called:
- a. Generalization
 - b. Specialization
 - c. Classification
 - d. Instantiation
-
69. In the COMPANY database, the number of locations per department suggests that LOCATION should be:
- a. A simple attribute of DEPARTMENT
 - b. A composite attribute
 - c. Modeled as a separate entity or multivalued attribute
 - d. Ignored in conceptual design
-
70. Which of the following is a very difficult conceptual point in ER modeling?
- a. Drawing rectangles for entities
 - b. Understanding that relationship attributes are necessary for M:N relationships
 - c. Knowing that SSN is a key
 - d. Recognizing that names can be composite
-

Answer Key – MCQs (1–70)

- 1. c
- 2. c
- 3. c
- 4. b
- 5. c
- 6. c
- 7. c
- 8. d

9. c
10. b
11. c
12. b
13. b
14. b
15. c
16. b
17. b
18. c
19. b
20. c
21. b
22. c
23. c
24. c
25. c
26. b
27. b
28. c
29. b
30. c
31. b
32. c
33. c
34. c
35. b
36. b
37. a
38. c
39. b
40. d
41. b
42. c
43. b
44. c
45. b
46. b
47. d

- 48. b
- 49. c
- 50. b
- 51. c
- 52. b
- 53. c
- 54. b
- 55. c
- 56. b
- 57. b
- 58. c
- 59. b
- 60. c
- 61. c
- 62. a
- 63. b
- 64. c
- 65. b
- 66. c
- 67. c
- 68. c
- 69. c
- 70. b

True/False Questions (30 Questions)

71. A simple attribute is atomic and cannot be subdivided.

- A. True
- B. False

72. In ER diagrams, all key attributes must be composite.

- A. True
- B. False

73. A weak entity can be uniquely identified without its owner entity.

- A. True

B. False

74. The relationship "SUPERVISION" is an example of a ternary relationship.

A. True

B. False

75. Multivalued attributes are represented by double ovals in ER diagrams.

A. True

B. False

76. The (min, max) notation (1,1) means mandatory and exclusive participation.

A. True

B. False

77. Every relationship type must have at least one attribute.

A. True

B. False

78. Total participation is graphically represented by a double line in ER diagrams.

A. True

B. False

79. The entity set is the same as the entity type.

A. True

B. False

80. A derived attribute is stored in the database.

A. True

B. False

81. In the COMPANY database, each project is controlled by exactly one department.

- A. True
 - B. False
-

82. An entity type may have more than one key, and all are underlined in ER diagrams.

- A. True
 - B. False
-

83. The attribute "HoursPerWeek" belongs to the EMPLOYEE entity.

- A. True
 - B. False
-

84. Recursive relationships require role names for clarity.

- A. True
 - B. False
-

85. The value set of an attribute defines its permissible values.

- A. True
 - B. False
-

86. A composite attribute cannot contain other composite attributes.

- A. True
 - B. False
-

87. The DEPENDENT entity is strong because it has a name attribute.

- A. True
 - B. False
-

88. In a 1:N relationship, relationship attributes can be moved to the N-side entity.

- A. True
- B. False

89. The degree of a relationship is the number of attributes it has.

- A. True
 - B. False
-

90. Existence dependency implies total participation.

- A. True
 - B. False
-

91. The MANAGES relationship connects EMPLOYEE and PROJECT.

- A. True
 - B. False
-

92. In ER modeling, a relationship type is a set of relationship instances.

- A. True
 - B. False
-

93. The partial key of a weak entity is underlined with a dashed line.

- A. True
 - B. False
-

94. UML Class Diagrams are never used for conceptual database design.

- A. True
 - B. False
-

95. The attribute SSN is a key attribute for EMPLOYEE in the COMPANY schema.

- A. True
 - B. False
-

96. A binary relationship involves exactly two entity instances per relationship instance.

- A. True

B. False

97. The WORKS_FOR relationship is many-to-many.

A. True

B. False

98. Every entity must participate in at least one relationship.

A. True

B. False

99. The start date of a department manager is an attribute of the DEPARTMENT entity.

A. True

B. False

100. In the (min, max) notation, max can be zero.

A. True

B. False

Answer Key – True/False (71–100)

71. A

72. B

73. B

74. B

75. A

76. A

77. B

78. A

79. B

80. B

81. A

82. A

83. B

84. A

- 85. A
 - 86. B
 - 87. B
 - 88. A
 - 89. B
 - 90. A
 - 91. B
 - 92. B
 - 93. B
 - 94. B
 - 95. A
 - 96. A
 - 97. B
 - 98. B
 - 99. B
 - 100. B
-